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Summary

Software Engineering Education researcher and educator with a strong focus on teaching excellence and student-centred learning. Experience includes teaching tertiary computing courses to large cohorts, designing adaptive learning systems, and leading large-scale educational data operations. Combines over 12 years of experience in higher education teaching, software development, learning analytics, and educational leadership.

Education

- 2023 – present **PhD Candidate in Computer Science, University of Alabama**
Advisor: Prof. Jeffrey Carver
Thesis: Learning Beyond the Lecture: Rethinking Software Engineering Education with Supplementary Materials in the Generative AI Era
Expected Completion: November 2026
- 2025 **MS in Computer Science, University of Alabama**
Comprehensive Exam: Effective Methods for Teaching Software Engineering
- 2023 **MSc by Research in Computing Science, University of the South Pacific**
Supervisor: Dr. Vani Vasudevan
Thesis: Intelligent Education Management Information System (IEMIS) for Fiji
- 2008 **BEd in Computer Science, University of Ibadan**
Minor: Mathematics
First Class (Honours); Best Graduating Student, Computer Science Department

Research in the Scholarship of Teaching and Learning (SoTL) & Computing Education

Journal Articles

- 1 L. Badru and J. Carver, "Personalizing supplementary materials in software engineering: Impact of preferences, learning styles, and gender," *ACM Transactions on Computing Education*, vol. 26, no. 4, 2026.
🔗 DOI: 10.1145/3806053
- 2 L. Badru, V. Vasudevan, M. Khan, and G. Lingam, "Mern stack web-based education management information systems for pacific island countries," *SN Computer Science*, vol. 4, no. 1, p. 70, 2022. 🔗 DOI: 10.1007/s42979-022-01437-0
- 3 L. Badru, V. Vasudevan, M. Khan, and G. Lingam, "State of education management information system (emis) in pacific melanesia small island developing states (sids)," *Directions: Journal of Educational Studies*, vol. 35, no. 2, pp. 107–118, 2021.

Conference Proceedings

- 1 L. Badru, M. An, J. Stamper, and J. Carver, "Optimizing learning: A comparative study of adaptive experiments and randomization in a software engineering course," in *Proceedings of the Sixth Annual Workshop on A/B Testing and Platform-Enabled Learning Engineering (PELE), Learning@Scale 2025*, Palermo, Italy, 2025.

Preprints

- 1 L. Badru and J. Carver, *How undergraduate software engineering students navigate learning materials across contexts*, Under Review *ACM Transactions on Computing Education (TOCE)*, Jun. 2026.

- 2 L. Badru and J. Carver, *What students use and what works: A systematic review in software engineering education*, Submitted to SIGCSE, Jul. 2026.

Work-in-progress

- 1 L. Badru and J. Carver, *Personalized support: Evaluating a digital platform for supplementary materials in software engineering education*, Work in progress, 2026.

Teaching Experience

- May, 2026  **Workshop Facilitator**, Tai Solarin Federal University of Education, Ijagun, Ogun State, Nigeria
- Designed and led a two-day virtual workshop on *Agentic Coding* for advanced Computer Science and Software Engineering students.
 - Created interactive activities and guided practice sessions to reinforce key concepts.
- 2016 – 2019  **Assistant Lecturer**, D.S. Adegbenro ICT Polytechnic, Ogun State, Nigeria
- Independently planned and delivered undergraduate computing, mathematics, and information systems courses for cohorts of up to 120 students.
 - Prepared lecture materials, tutorials, laboratory exercises, and assessment tasks aligned with the NBTE-approved curriculum.
 - Sequenced course topics and adapted the pace of instruction to improve student understanding while maintaining curriculum requirements.
 - Marked assignments, practical work, and examinations, providing constructive feedback to support student learning.
 - Provided academic mentoring and informal supervision for numerous undergraduate student projects.
- 2014 – 2016  **Education Officer**, Staff Development Center, Bureau of Establishment & Training, Nigeria
- Delivered training in computer applications and digital literacy for adult learners and administrative staff.
 - Developed instructional materials and hands-on exercises tailored to diverse learner backgrounds.
- 2010 – 2026  **Mentorship and Student Support**
- Mentored over 100 high school and college students, supporting academic growth and career development.
 - Provided guidance on computing fundamentals, project work, and professional pathways.

Employment History

2023 – present	Research Assistant, Department of Computer Science, University of Alabama • Conduct research in software engineering education and learning analytics. • Designed adaptive learning experiments used across multiple universities • Contributed to ACM publications and data analysis workflows
2016 – 2022	Senior Education Officer / Acting Head, State EMIS, Ministry of Education, Science and Technology, Abeokuta, Nigeria • Directed statewide EMIS operations supporting 1,500+ schools and 500,000+ students • Developed analytics tools that improved reporting accuracy and reduced processing time • Trained 1,000+ personnel as UNICEF Master Trainer • Contributed to Education Sector Plan (2017–2030) • Served as a technical expert for the Education Policy & Strategy Simulation (EPSSim) initiative.
2016 – 2019	Assistant Lecturer (Secondment), D.S. Adegbenro ICT Polytechnic, Nigeria • Acted as Course Authority with full autonomy over curriculum sequencing, assessment design, and final grade moderation for cohorts up to 120 students • Developed comprehensive tutorials, lecture notes, and practical laboratory exercises. • Facilitated student mastery of complex topics using structured lessons and step-by-step worked examples.
2014 – 2016	Education Officer (Secondment), Staff Development Center, Bureau of Establishment & Training, Ogun State, Nigeria • Delivered training in computer applications and digital literacy for confidential secretaries and administrative staff.
2010 – 2014	Programmer Analyst, Teaching Service Commission, Abeokuta, Nigeria • Provided technical support for educational data systems and contributed to ICT capacity-building initiatives.

Skills

Programming Languages	Python, JavaScript, PHP, React, Node.js
Artificial Intelligence	OpenAI API, Prompt Engineering, AI-driven Adaptive Learning Systems
Learning Analytics	Personalization Models, Predictive Analytics, Behaviour Modelling, A/B Testing
Educational Technology	Adaptive Experimentation Platforms, Learning Tool Integration, Technology-enhanced Teaching
Learning Delivery Platforms	Moodle, Blackboard, Canvas, Open Learning Initiative (OLI)
Research Methods	Experimental Design, Mixed Methods, Survey Design, Statistical Modelling

Skills (continued)

Data & Research Tools	■ Alteryx, Tableau, Pandas, NumPy, SciPy
Databases	■ MySQL, PostgreSQL, MongoDB
Software Engineering Education	■ Curriculum Design, Instructional Materials, Pedagogical Strategies, Learning Resource Evaluation
Teaching & Facilitation	■ Lecture Design, Assessment Development, Worked Examples, Project-based Learning, Large-class Teaching

Miscellaneous Experience

Scholarly Contribution

- 2026 ■ **Reviewer**, *Special Interest Group on Computer Science Education (SIGCSE)*. Review manuscripts in Computing Education Research (CER).
- 2022 ■ **Reviewer**, *Education and Information Technologies* (Springer Nature). Reviewed manuscripts in educational technology and learning analytics.

Selected Industry & Development Projects

- 2026–present ■ **Security Geolocation Application**. Developing a security-focused geolocation application to support operational decision-making in Nigeria.
 - Implementing real-time location tracking, mapping, and secure data management.
 - Applying software engineering principles across requirements, architecture, and iterative development.
 - Building practical, location-aware tools for high-stakes public safety contexts.
- 2026 ■ **Financial Management Information System (FMIS)**. Co-developed a web-based FMIS for thrift and cooperative societies in Nigeria.
 - Contributed to requirements analysis, system design, database modelling, and iterative development.
 - Strengthened practical experience in secure data handling and operational software systems.

Awards and Scholarships

- 2021 ■ **Research Grant**, Queen Elizabeth Commonwealth Scholarship, United Kingdom (3,000 USD)
- 2020 – 2021 ■ **Commonwealth Scholarship**, United Kingdom (value over 50,000 USD), University of the South Pacific, Fiji
- 2019 ■ **Dean's Award for Academic Excellence & Global Leader Scholarship**, University of Bath, United Kingdom (value £10,000)
- 2008 ■ **Best Graduating Student**, Department of Computer Science, Tai Solarin University of Education, Nigeria (value 500 USD)

Certifications

- 2021 ■ **Predictive Analytics for Business**, Udacity (Alteryx & Tableau)
- **Microsoft Azure Fundamentals**, Microsoft
- **Ideas for a Better World: Leading Change Through Policymaking**, British Council

Miscellaneous Experience (continued)

Mentorship

- 2010 – 2026  Advised and mentored over 100 senior secondary and university students, supporting academic and professional development.
- 2026  **Judge**, Alabama Robotics Contest, University of Alabama. Evaluated robot navigation accuracy, task completion, and engineering design.

Professional Memberships

-  **Association for Computing Machinery (ACM)**
-  **Institute of Electrical and Electronic Engineers (IEEE)**
-  **Special Interest Group on Computer Science Education (SIGCSE)**

References

Available on Request